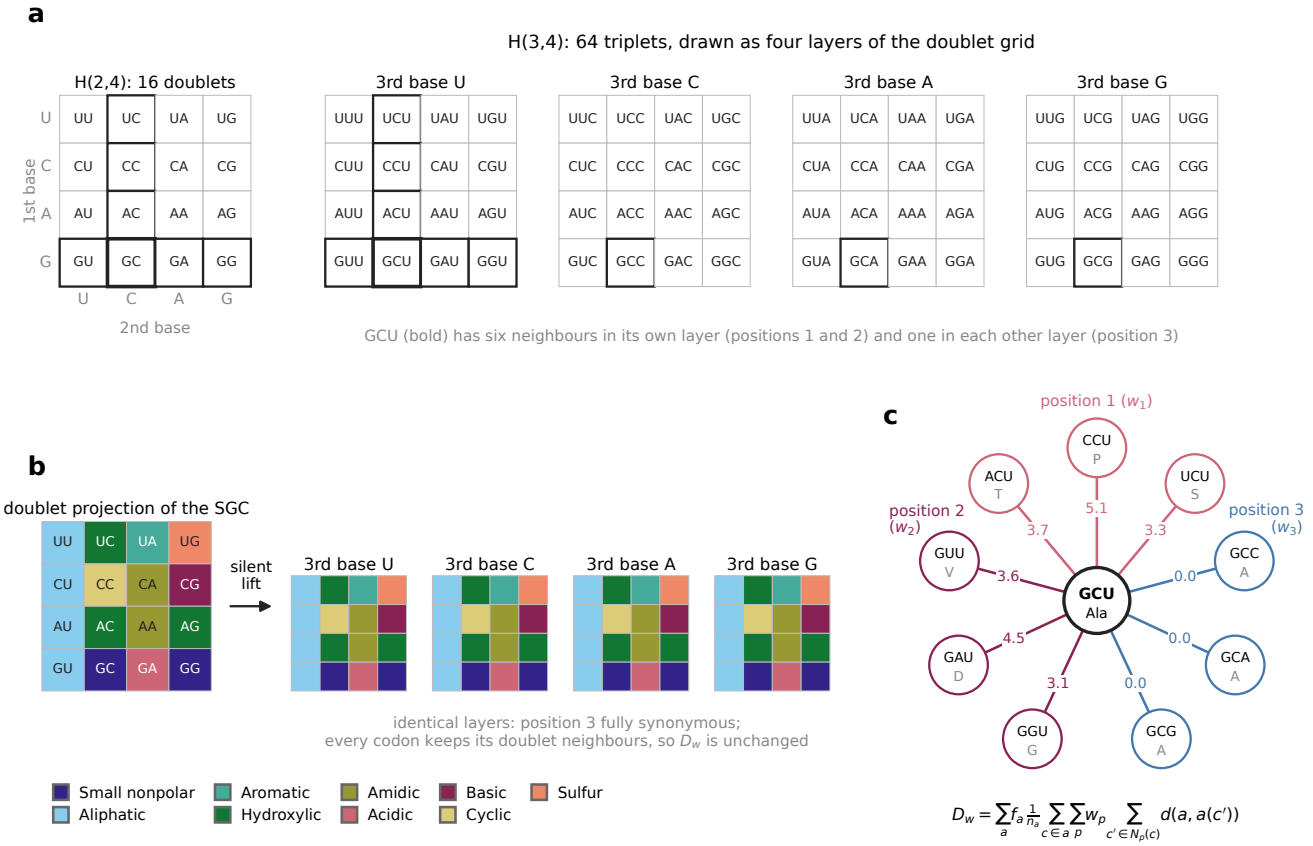


Figure 1. Codes on the Hamming graph and the silent lift. (a) $H(2, 4)$ drawn as a 4 × 4 grid (rows, first base; columns, second base), and $H(3, 4)$ drawn as four copies of that grid, one per third base (after Lenstra 2015). Two codons are adjacent when they differ at one position: GCU (bold outline) has six neighbours in its own layer, at positions 1 and 2, and one neighbour in each of the other three layers, at position 3. (b) The doublet projection of the SGC (left; nine classes, coloured by class) and its silent lift (right): every layer of the triplet code is a copy of the doublet code, so position 3 is fully synonymous and every codon keeps the non-synonymous neighbours it had as a doublet. (c) The neighbourhood of GCU (Ala) in the SGC, with the physicochemical distance d to each neighbouring amino acid; position-3 neighbours are synonymous ($d = 0$). The per-residue cost D_W sums these distances over all neighbours, weighting each position by its misreading rate w_p and each amino acid by its usage f_a.

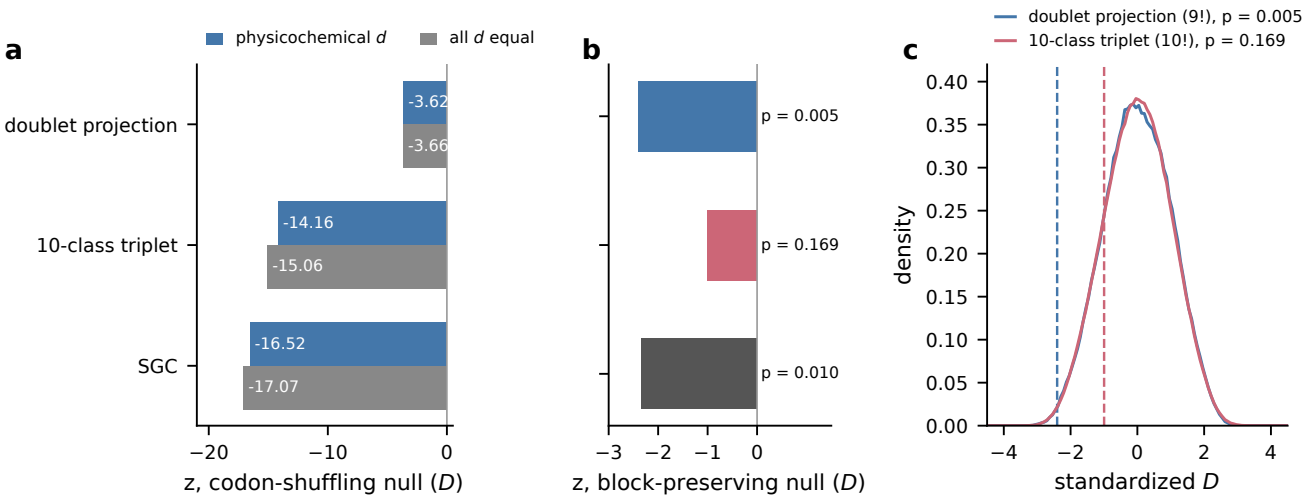


Figure 2. Codon-shuffling nulls measure synonym clustering. (a) z-scores of the noise distortion D of the doublet projection, the ten-class triplet projection and the SGC against the codon-shuffling null (10^6 shuffles, seed 20260924), with physicochemical distances (blue) and with every pair of amino acids at the same distance (grey). (b) The same codes against the block-preserving null; the two projections were enumerated exhaustively (9! and 10! codes) and the SGC was sampled 10^7 times. p is the fraction of codes strictly better than the observed code. (c) Exhaustive block-preserving null distributions of D for the two projections, standardized to zero mean and unit variance; dashed lines mark the observed codes.

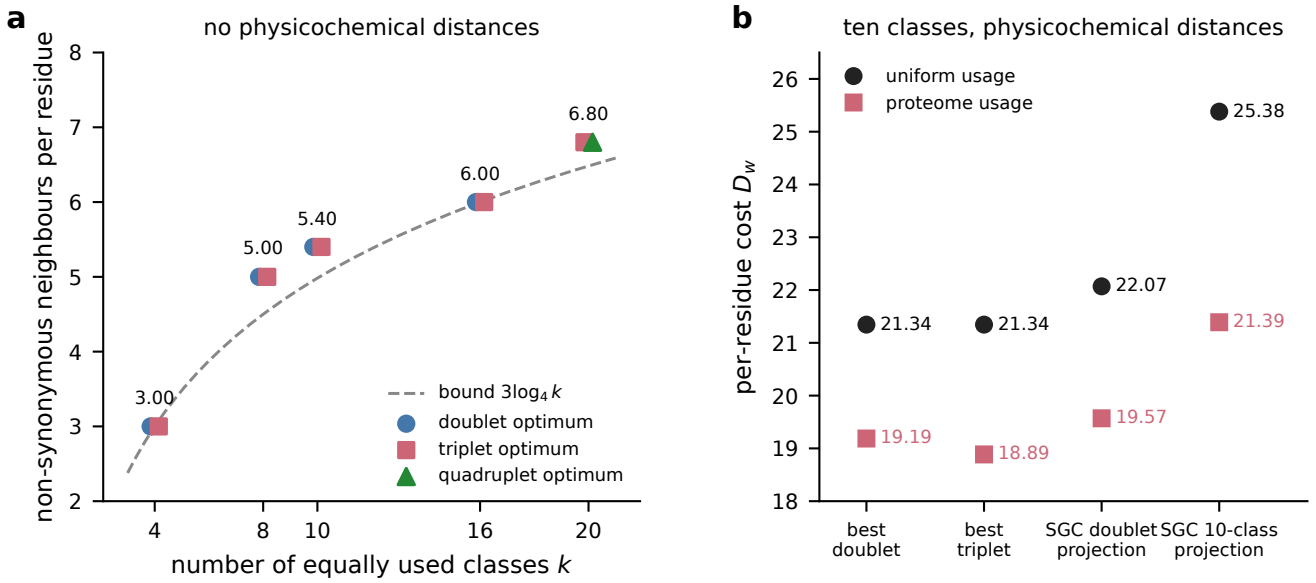


Figure 3. Codon length does not change the achievable cost. (a) Minimum number of non-synonymous neighbours per residue, without physicochemical distances, for doublets (circles) and triplets (squares) with k equally used classes, and for triplets and quadruplets with 20 classes (k = 20). The dashed line is the edge-isoperimetric bound 3 $\log_4 k$, attained at k = 4 and 16. (b) Minimum per-residue cost D_W for ten classes with physicochemical distances under uniform (circles) and proteome (squares) usage, for the best doublet and triplet codes and for the two projections of the SGC.

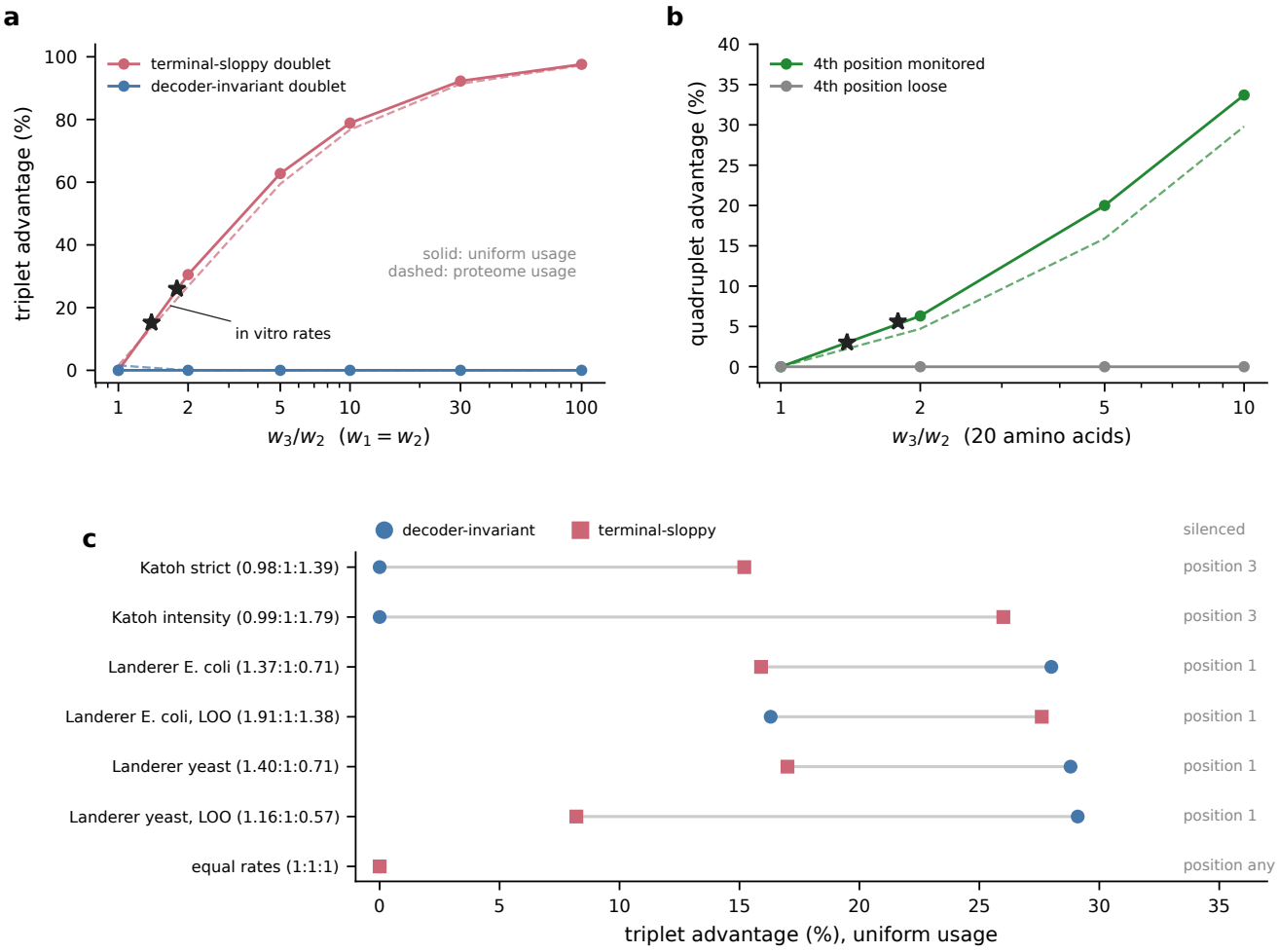


Figure 4. The advantage of a longer code is the cost of the channel it discards. (a) Advantage of the best triplet code over the best doublet code, $1 - D_3^*/D_2^*$, against w_3/w_2 (with $w_1 = w_2$), under the terminal-sloppy and decoder-invariant assumptions; solid lines, uniform usage; dashed lines, proteome usage. Stars mark the in vitro misreading rates of Katoh and Suga (2023). (b) Advantage of the best quadruplet code over the best triplet code for 20 amino acids, when the fourth position is read as accurately as position 2 (monitored) or at the position-3 rate (loose); stars mark the in vitro rates. (c) Triplet advantage at the seven published rate sets of table 2 under the two assumptions about doublet decoding (uniform usage), with the position the best triplet code silences.

Supplementary Materials

Companion to “Error robustness of doublet and triplet genetic codes”
Hannah E. Ribbeck, J. Paudyal, Boggavarapu Kiran
Submitted to the Journal of Molecular Evolution (revised version).

Table S1. Sensitivity to the representative amino acid of each class

Five classes have two members equidistant from the class centroid ({A, G}, {S, T}, {N, Q}, {D, E}, {K, R}), so the representative is arbitrary. The table gives, for all 32 choices, the noise distortion *D* of the doublet projection (A, nine classes) and of the ten-class triplet projection (C), with the exhaustive block-preserving p-value (fraction of the 9! or 10! codes strictly better). The four larger classes have fixed representatives (L, W, H, P, C). Rebuilt descriptor matrix. The first row is the choice used in the main text.

Representatives (X1 to X10)	*D* (A)	p (A, 9!)	*D* (C)	p (C, 10!)
ALWSNDRHPC	18.794	0.0051	21.906	0.169
ALWSNDKHPC	18.224	0.0017	20.875	0.043
ALWSNEKHPC	18.521	0.0002	20.970	0.012
ALWSNERHPC	19.090	0.0007	22.048	0.090
ALWSQDKHPC	18.327	0.0016	20.751	0.027
ALWSQDRHPC	18.960	0.0047	21.916	0.152
ALWSQEKHPC	18.407	0.0001	20.733	0.009
ALWSQERHPC	19.040	0.0003	21.944	0.081
ALWTNDKHPC	18.489	0.0060	21.215	0.071
ALWTNDRHPC	19.025	0.0129	22.159	0.196
ALWTNEKHPC	18.786	0.0016	21.310	0.028
ALWTNERHPC	19.322	0.0040	22.301	0.107
ALWTQDKHPC	18.481	0.0075	21.033	0.057
ALWTQDRHPC	19.081	0.0139	22.111	0.183
ALWTQEKHPC	18.561	0.0015	21.015	0.025
ALWTQERHPC	19.161	0.0033	22.139	0.105
GLWSNDKHPC	18.039	0.0020	20.822	0.051
GLWSNDRHPC	18.438	0.0042	21.585	0.145
GLWSNEKHPC	18.635	0.0025	21.231	0.035
GLWSNERHPC	19.034	0.0032	22.041	0.112
GLWSQDKHPC	18.141	0.0004	20.698	0.016
GLWSQDRHPC	18.604	0.0007	21.595	0.087
GLWSQEKHPC	18.521	0.0003	20.993	0.011
GLWSQERHPC	18.984	0.0004	21.937	0.063
GLWTNDKHPC	18.496	0.0109	21.330	0.100
GLWTNDRHPC	18.863	0.0169	22.007	0.200
GLWTNEKHPC	19.093	0.0133	21.740	0.074
GLWTNERHPC	19.459	0.0141	22.463	0.156
GLWTQDKHPC	18.488	0.0048	21.148	0.051
GLWTQDRHPC	18.918	0.0072	21.958	0.138
GLWTQEKHPC	18.868	0.0044	21.444	0.038
GLWTQERHPC	19.298	0.0046	22.301	0.105

Across the 32 choices, p for the doublet projection ranges from 0.0001 to 0.0169 and p for the ten-class triplet projection from 0.009 to 0.200.

Doublet projection with ten classes. The majority vote leaves four ties (AA, AG, CA, UU). The main text breaks them by the order of the class identifiers X1 to X10, which gives CA to Asn/Gln and leaves His without a prefix. Ordering by amino-acid letter instead gives CA to His but AA to Lys/Arg, so Asn/Gln loses both of its prefixes and the projection still has nine classes. All ten classes survive only if each tie goes to the class that would otherwise lose its last prefix (AA to Asn/Gln, AG to Ser/Thr, CA to His, UU to the aliphatic class). With the main-text representatives that projection has $D = 19.974$, and the exhaustive block-preserving null over 10! codes gives p = 0.0430, against 0.169 for the ten-class triplet projection.

Table S2. Amino-acid descriptors

Twenty-two side-chain properties were targeted. Each descriptor is an AAindex1 entry (file downloaded 2026-09-24; SHA-256 560f102d…f676) or a deterministic structural rule computed with RDKit 2026.03.1 on the SMILES of the free amino acid. The rules for choosing entries were fixed, and their file hashed (SHA-256 36802edb…688a), before any statistic was computed. No pair of retained descriptors correlates at $|r| \geq 0.95$ (largest $|r| = 0.943$).

Property	Source	Accession or rule	Title	Status
hydropathy	AAindex1	KYTJ820101	Hydropathy index (Kyte-Doolittle, 1982)	kept
volume	AAindex1	BIGC670101	Residue volume (Bigelow, 1967)	kept
polarity	AAindex1	GRAR740102	Polarity (Grantham, 1974)	kept
isoelectric point	AAindex1	ZIMJ680104	Isoelectric point (Zimmerman et al., 1968)	kept
side-chain pKa	none			dropped: no AAindex entry (FASG760104, FASG760105, FAUJ880113 and JOND750102 describe the alpha-amino and alpha-carboxyl groups); no structural rule
net charge	AAindex1	KLEP840101	Net charge (Klein et al., 1984)	kept
hydrogen-bond donors	AAindex1	FAUJ880109	Number of hydrogen bond donors (Fauchère et al., 1988)	kept
hydrogen-bond acceptors	RDKit	CalcNumHBA (free amino acid)	RDKit count	kept
aromaticity	RDKit	1 if any aromatic atom else 0	RDKit indicator	kept
ring count	RDKit	CalcNumRings (free amino acid)	RDKit count	kept
rotatable bonds	RDKit	CalcNumRotatableBonds (free amino acid)	RDKit count	kept
molar refractivity	AAindex1	MCMT640101	Refractivity	kept

Property	Source	Accession or rule	Title	Status
			(McMeekin et al., 1964)	
bulkiness	AAindex1	ZIMJ680102	Bulkiness (Zimmerman et al., 1968)	kept
helix propensity	AAindex1	CHOP780201	Normalized frequency of alpha-helix (Chou-Fasman, 1978)	kept
strand propensity	AAindex1	CHOP780202	Normalized frequency of beta-sheet (Chou-Fasman, 1978)	kept
turn propensity	AAindex1	CHOP780203	Normalized frequency of beta-turn (Chou-Fasman, 1978)	kept
solvent accessibility	AAindex1	ROSG850102	Mean fractional area loss (Rose et al., 1985)	kept
flexibility	AAindex1	VINM940101	Normalized flexibility parameters, average (Vihinen et al., 1994)	kept
hydrophobic moment	AAindex1	EISD860102	Atom-based hydrophobic moment (Eisenberg-McLachlan, 1986)	kept
strand moment	none			dropped: no AAindex entry and no structural rule
hydrophobic surface	AAindex1	ROSG850101	Mean area buried on transfer (Rose et al., 1985)	kept
side-chain mass	AAindex1	FASG760101	Molecular weight (Fasman, 1976)	kept

Table S3. Sensitivity to the descriptor matrix

Main statistics under four representations: the rebuilt matrix (table S2) with each of the eight principal components rescaled to unit variance (primary) or unscaled, and the earlier matrix compiled by hand from published scales, rescaled or unscaled. A, doublet projection; B, SGC; C, ten-class triplet projection. Codon-shuffling null: 10^6 draws, seed 20260924, identical draws for every matrix. Block-preserving null: A and C exhaustive, B 10^7 draws.

Statistic	Rebuilt, rescaled (primary)	Rebuilt, unscaled	Earlier, rescaled	Earlier, unscaled
Codon-shuffling z, *E*, A	-2.805	-2.169	-3.319	-2.245
Codon-shuffling z, *D*, A	-3.619	-3.144	-3.798	-3.104
Codon-shuffling z, *E*, B	-13.949	-11.968	-14.603	-12.816
Codon-shuffling z, *D*, B	-16.523	-15.243	-16.563	-15.560
Codon-shuffling z, *E*, C	-11.901	-11.112	-13.349	-11.610
Codon-shuffling z, *D*, C	-14.160	-13.498	-14.811	-13.803
Block-preserving p, *E*,	0.007	0.085	0.007	0.193

Statistic	Rebuilt, rescaled (primary)	Rebuilt, unscaled	Earlier, rescaled	Earlier, unscaled
A				
Block-preserving p, *D*, A	0.005	0.065	0.007	0.175
Block-preserving p, *E*, B	0.027	0.006	0.023	0.011
Block-preserving p, *D*, B	0.010	0.001	0.005	0.001
Block-preserving p, *E*, C	0.171	0.525	0.319	0.668
Block-preserving p, *D*, C	0.169	0.510	0.340	0.671

With all pairwise distances equal, the codon-shuffling z is −3.660 (A), −17.072 (B) and −15.057 (C) for every matrix.

Correlation threshold. Lowering the pruning threshold from $|r| \geq 0.95$ to 0.94 removes net charge ($|r| = 0.941$ with isoelectric point) and flexibility ($|r| = 0.943$ with mean fractional area loss); eight components still carry 97.5% of the variance. Every statistic reaches the same verdict:

Statistic	Threshold 0.95 (primary)	Threshold 0.94
Codon-shuffling z, *D*, A / B / C	−3.62 / −16.52 / −14.16	−3.74 / −16.64 / −14.22
Block-preserving p, *D*, A (9!)	0.0051	0.0039
Block-preserving p, *D*, C (10!)	0.169	0.130
Block-preserving p, *D*, B (10^6 draws)	0.0103	0.0090
Ten classes, uniform usage: doublet optimum / triplet optimum	21.345 / 21.345	21.565 / 21.565

Table S4. Position-specific misreading rates

Relative rates $w_1 : w_2 : w_3$ (position 2 = 1) per non-synonymous opportunity. Every assay sees only amino-acid-changing misreading, so w_3 is conditional on the position-3 change altering the amino acid.

Source	System	$w_1 : w_2 : w_3$	95% interval	Single-mismatch events (pos. 1 / 2 / 3)	Notes
Katoh and Suga 2023, Suppl. Fig. 8, strict	*E. coli*, reconstituted translation, cognate amino acid withheld	0.98 : 1 : 1.39	w_1/w_2 0.69–1.40; w_3/w_2 0.95–2.05	combinations tested 64 / 71 / 21; positive 30 / 34 / 14	fraction of single-mismatch codon–amino-acid combinations with detectable misincorporation
Katoh and Suga 2023, intensity-weighted	as above	0.99 : 1 : 1.79	—	as above	mean of the 0–5 intensity score; ordinal, indicative only
Katoh and Suga 2023, authors' one-mismatch class	as above	0.94 : 1 : 1.39	w_1/w_2 0.65–1.36; w_3/w_2 0.95–2.05	62 / 71 / 21	differs from strict by two combinations whose printed totals disagree with their mismatch rows
Landerer et al. 2024, Data S4	*E. coli*, 80 proteomes in vivo	1.37 : 1 : 0.71	dataset bootstrap: w_1/w_2 1.10–2.55; w_3/w_2 0.32–2.25	13,029 / 10,505 / 2,157 PSMs (of 58,906)	events assigned to the position at which a codon of the new amino

Source	System	$w_1 : w_2 : w_3$	95% interval	Single-mismatch events (pos. 1 / 2 / 3)	Notes
					acid differs from the original codon; rate = events / (codon exposure × detectable non-synonymous changes at that position)
Landerer et al. 2024, largest dataset (PXD014341) excluded	*E. coli*	1.91 : 1 : 1.38	—	—	PXD014341 holds 47% of single-mismatch PSMs
Landerer et al. 2024, cysteine origins removed	*E. coli*	1.22 : 1 : 0.71	—	—	Cys-origin events are candidate preparation artefacts
Landerer et al. 2024, Data S4	yeast, 72 proteomes in vivo	1.40 : 1 : 0.71	dataset bootstrap: w_1/w_2 0.76–2.54; w_3/w_2 0.35–1.19	3,106 / 2,470 / 636 PSMs (of 17,765)	as above
Landerer et al. 2024, largest dataset (PXD026077) excluded	yeast	1.16 : 1 : 0.57	—	—	PXD026077 holds 11% of PSMs
Landerer et al. 2024, cysteine origins removed	yeast	1.39 : 1 : 0.71	—	—	

Sources not used: Kramer and Farabaugh (2007, Table 1) lists only codons read above background for one tRNA, so per-position means cannot be computed; Mordret et al. (2019) and Manickam et al. (2016) report position data only in figures, which we did not digitize.

Table S5. Ten-class optima under position weights (w_1, 1, w_3)

Rebuilt descriptor matrix, free class sizes, all codons sense. D_2*(inv), best doublet read with weights (w_1, 1); D_2*(term), best doublet read with weights (w_1, w_3); D_3*, best triplet. A, triplet advantage $1 - D_3/D_2$*. Every triplet value was checked against all three single-position silent lifts. Silenced: codon position made fully synonymous in the best triplet code (none, the best code beats every lift).

Usage	w_1	w_3	D_2*(inv)	D_2*(term)	D_3*	A, decoder-invariant	A, terminal-sloppy	Silenced
uniform	1	1	21.345	21.345	21.345	0.0%	0.0%	3
uniform	1	2	21.345	30.719	21.345	0.0%	30.5%	3
uniform	1	5	21.345	57.318	21.345	0.0%	62.8%	3
uniform	1	10	21.345	100.988	21.345	0.0%	78.9%	3
uniform	1	30	21.345	275.667	21.345	0.0%	92.3%	3
uniform	1	100	21.345	887.045	21.345	0.0%	97.6%	3
uniform	3	1	39.638	39.638	21.345	46.2%	46.2%	1
uniform	3	2	39.638	52.309	30.719	22.5%	41.3%	1
uniform	3	5	39.638	83.055	39.638	0.0%	52.3%	3

Usage	w_1	w_3	D_2*(inv)	D_2*(term)	D_3*	A, decoder-invariant	A, terminal-sloppy	Silenced
uniform	3	10	39.638	127.761	39.638	0.0%	69.0%	3
uniform	3	30	39.638	302.963	39.638	0.0%	86.9%	3
uniform	3	100	39.638	914.341	39.638	0.0%	95.7%	3
proteome	1	1	19.187	19.187	18.886	1.6%	1.6%	none
proteome	1	2	19.187	26.221	19.187	0.0%	26.8%	3
proteome	1	5	19.187	47.323	19.187	0.0%	59.5%	3
proteome	1	10	19.187	82.493	19.187	0.0%	76.7%	3
proteome	1	30	19.187	222.493	19.187	0.0%	91.4%	3
proteome	1	100	19.187	711.945	19.187	0.0%	97.3%	3
proteome	3	1	33.255	33.255	19.187	42.3%	42.3%	1
proteome	3	2	33.255	45.409	26.082	21.6%	42.6%	none
proteome	3	5	33.255	71.630	33.255	0.0%	53.6%	3
proteome	3	10	33.255	106.800	33.255	0.0%	68.9%	3
proteome	3	30	33.255	247.478	33.255	0.0%	86.6%	3
proteome	3	100	33.255	737.402	33.255	0.0%	95.5%	3